Red Team Critique of Blue Team Version Three

Version three makes real progress. It gives a worked example, names private set intersection, separates merchant-facing from reviewer-facing explanations, and moves toward signals based on counterparties. But the formula it presents as mathematical rigor contradicts its own example, several cited faults are not the ones it answers, and most new features fail on the honest merchants the system is meant to protect.

Critical faults

1. The capacity formula contradicts its own example.
Version three divides total alerts by target recall and then divides again by precision. Alerts already depend on precision, so the same effect is counted twice. Applying the formula to the example's own figures, 12,857 alerts divided by 0.9 and then by 0.14 gives about 102,000 reviews, roughly eight times the 12,857 the example reports. The correct relationship is reviews needed equal abusers multiplied by target recall, divided by precision, which is 2,000 times 0.9 divided by 0.14, or about 12,857. Three further problems remain:

- The example calls those reviews "confirmed", but only 1,800 are confirmed and the rest are false alarms.
- There is no analyst throughput. At an assumed twenty reviews per analyst per working day over about sixty-five working days, the quarter needs roughly ten analysts, plus legal reviewers that version three never counts.
- The 14 percent figure was the minimum precision that fit one capacity assumption, not a target. Whether a model can reach 90 percent recall at that precision depends on a measured precision and recall curve, which needs labels that version three still has no plan to collect.

2. Several faults are cited but never answered.
Section one cites fault twelve, but the second critique had only eleven. Section three cites faults three and five, which concerned the label feedback loop and the contamination of anchors and cohorts, yet the section covers account control and privacy. None of the following receives an answer:

- Random audits of unflagged merchants and a protected holdout set.
- Cumulative change tests for slow drift. The Kalman filter has simply been dropped.
- Cohorts that absorb widespread abuse, and the absence of a process for re-declaring a business profile.
- Tier gaming, scale and cost estimates, and insider risk.

3. The clearing path and the randomization both create new problems.

- Automatic clearing after thirty days without drift gives abusers a recipe: behave normally for thirty days. It also relies on a "sanity check" against static risk factors, which is the document checking the first critique found insufficient.
- Clearing Green merchants does not shorten the review queue, because they were never in it. The claim of no loss in recall is also beside the point, since recall covers all abusers, including those cleared as Green.
- Randomization was suggested earlier, but version three applies it to velocity thresholds and cohort boundaries without limits. Similar merchants could be treated differently by design, which weakens fairness, the audit trail, and the explanations in section four. Randomness belongs in audit sampling and hidden secondary checks, within documented bounds, with tests that treatment stays consistent.

4. The Flow Divergence Index is incoherent as defined.
Net outflow to external entities divided by total volume has two readings, and both fail. If net means outflow minus inflow, a true pass-through account scores near zero and is labelled legitimate. If net means gross outflow, every merchant that pays suppliers, wages, rent, or refunds scores high. Wholesalers, marketplaces, travel agents, landlords, and payroll services also score high, and there is no cohort adjustment. The real signature of pass-through is dwell time: how long money stays in the account, what share of inflow leaves within a short window, and how many distinct payers were involved.

5. The account-control signals bring back the weak signals from version two.
Frequent changes of login network address and virtual private network use were exactly what the previous critique called unreliable, because honest owners buy phones, travel, share mobile network addresses, and use such networks for privacy. Version three lists them again as "the true signal", and it drops the problem of sole proprietors settling to personal accounts. Detecting delegation "through shared transaction patterns" and comparing the operator's claimed identity with payers' identities are asserted without a method, and a provider seldom knows who its merchants' payers really are. Rented accounts are better revealed by where logins occur compared with the shop's address and opening hours, whether the owner's own devices still appear, who initiates payouts, and whether payers live far from the shop.

Major faults

6. Private set intersection is described as something it is not.
Exchanging cryptographic hashes of identifiers is not private set intersection. Phone numbers, identity numbers, and network addresses come from small search spaces, so hashes can be reversed by trying every candidate. Real protocols, such as those built on oblivious pseudorandom functions, reveal only the matches. Further problems:

- "Transaction fingerprints" cannot link providers, because each provider sees different transactions.

- Hashed network address ranges link unrelated merchants on shared mobile networks.
- The output of such a protocol is an intersection, not a risk score, so a second computation is needed.
- The mention of a "central system" adds a party the design never describes.
- Governance remains open: who runs the consortium, what evidence is required before a match is shared, how a wrong match is corrected, who is liable, and how scores from different providers are calibrated.

7. The betting features are still partly wrong and miss the strongest signal.

- The Payer Concentration Index is defined as concentration of funds flowing to a single recipient, which is the merchant side of the flow. The betting signature is many distinct payers each paying repeatedly. If concentration means a few payers dominate, it points the wrong way. "Distance in time and space between payers" is undefined, and payer location may not exist for every payment method.
- Repeat payments alone also match tuition, subscriptions, rent, and parking. They must be combined with stake-like amounts and intraday timing.
- Small, frequent disbursements to distinct accounts also describe wages, refunds, and cashback. The distinguishing feature is a closed loop: people paid out were earlier payers, with amounts and timing tied to their own payments. Refunds must be separated by size and timing.
- Payouts made in cash or through other channels remain invisible. The event calendar still covers sports and holidays only, and all-day games and the stadium vendor overlap remain unresolved.

8. The legal triage layer asks software to do legal work.
An automated layer that "validates the risk profile against jurisdictional rules" needs someone to write, maintain, and answer for those rules. Recording "the exact legal rationale for any action" in the analyst view assigns to a system what only counsel can supply. Sending every critical case to legal review adds a second capacity limit that the formula ignores, since lawyers are scarcer than analysts. Because no restrictive action may occur before triage, money keeps moving while cases wait. A graded approach would solve this: reversible, low-impact measures such as closer monitoring, document requests, and settlement delay within contract terms at analyst level, and freezes only after legal review.

9. Due process protections shrank, and the merchant explanation still leaks.
"Temporary Hold due to Payer Flow Anomaly" tells an abuser which signal family fired. "Account under review" may itself fall under rules against tipping off in some jurisdictions, and version three asserts compliance without naming a rule or requiring counsel to approve standard wording. The appeal path, the time limit on holds, and the treatment of held funds from version two have vanished. Measuring the time between flag and review is a metric, not a limit.

What Blue Team version four should fix

Fault	Required fix
Capacity formula	Use reviews equal abusers times recall divided by precision, then add analyst throughput and legal reviewer capacity
Unanswered faults	Add random audits, a protected holdout set, cumulative change tests, cohort re-declaration, tier gaming defenses, cost estimates, and insider risk controls
Automatic clearing	Replace thirty-day clearing with continued low-cost monitoring and random audits of cleared merchants
Randomization	Bound it, document it, and test that similar merchants receive consistent treatment
Flow index	Define it with dwell time, share of inflow leaving within a short window, and payer diversity, relative to the cohort
Account control	Use login location against shop address and hours, device history, and payout initiator, and treat delegation as normal
Linkage	Use a real private set intersection protocol and name governance, liability, and correction
Betting features	Match payouts to earlier payers, combine repeat rate with stake-like amounts, and cover all-day games
Legal work	Have counsel write the rules and grade actions by reversibility
Due process	Restore appeal paths, time limits, and rules for held funds, with counsel-approved neutral wording

Bottom line: version three improved the vocabulary again, and its worked example was a step forward, but the formula fails its own check and the design retreated on due process. The strongest next version would fix the arithmetic first and close the label problem with random audits, since every other number depends on labels. It would also define each feature precisely enough that someone could compute it.

I can draft Blue Team version four on that basis, or put this critique into a document if you would like.